Communitree

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How do we Address Community Disconnection?

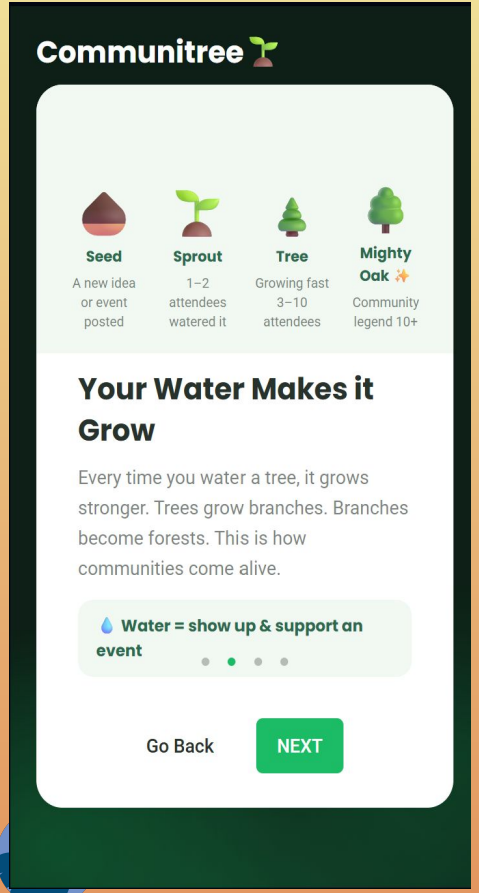
Rising Social Isolation

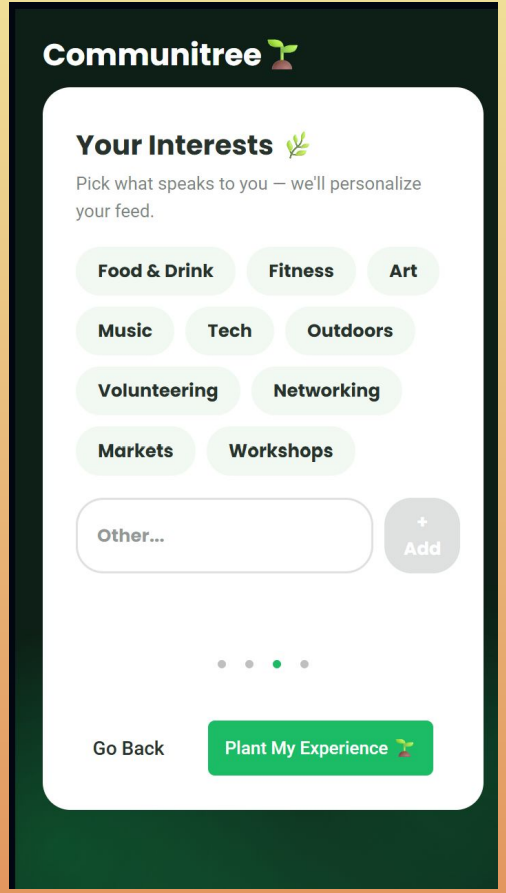
- Social isolation increased by 13.4% over a 16-years, the entire increase occurring after 2019.

Event Discovery is Broken

- 60–70% of people rely on social media or word of mouth to find local events
- Events are fragmented across social platforms
- Users miss events because there is no central hub

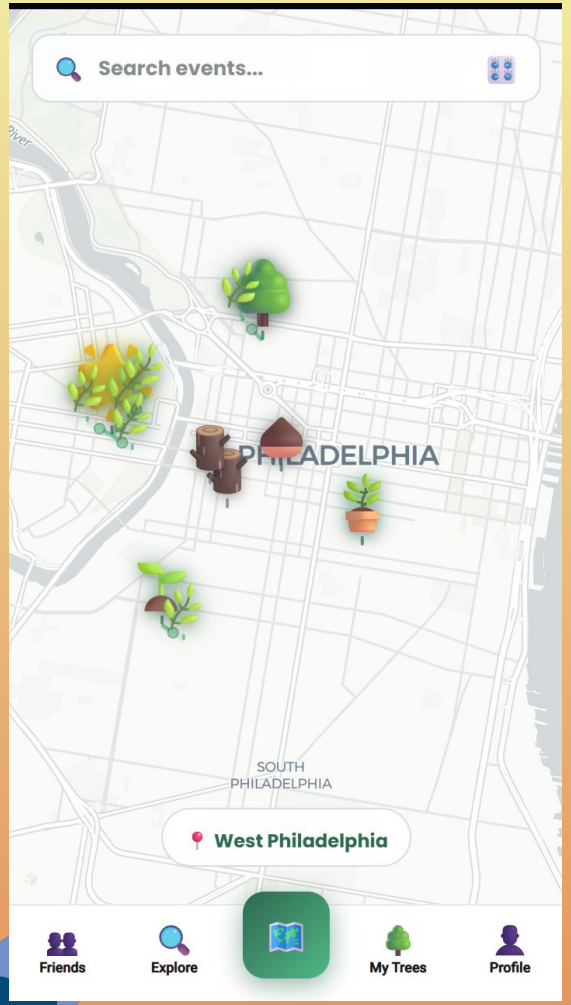
How do we use Communitree?





We help you plant your communitree

- We connect your different interest together based on **location** and **interest surveys**
- Foster community engagement through events!

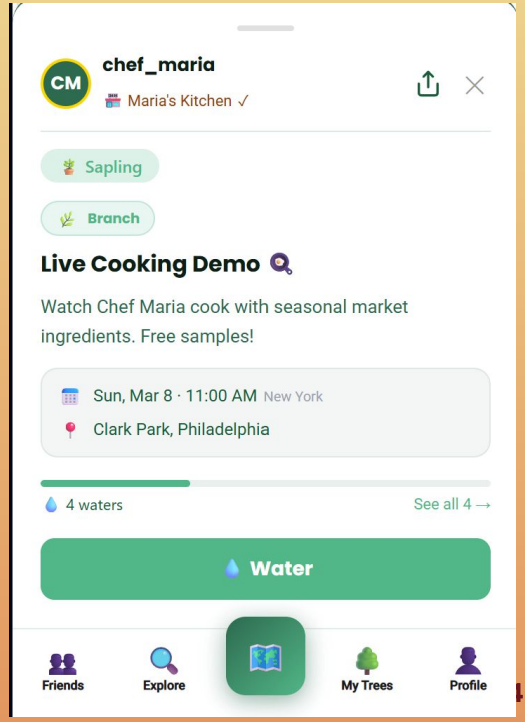


Discover Your Community!

Start organizing communal based events and invite both friends and the community!

What is **WATERING,**

- You can like certain events and help those events grow BIGGER
- More water, the more advertising we do for it on our regional map!



How do I Track the Events I Planted/water?

Planting

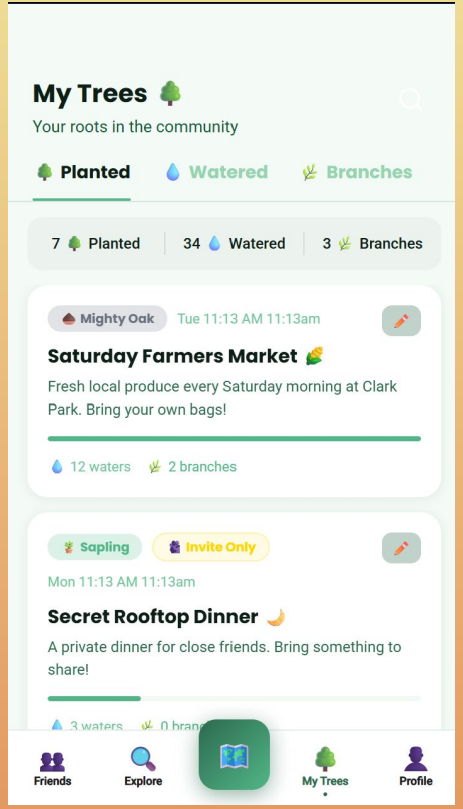
- Planting seeds to publish events
- Public or private events

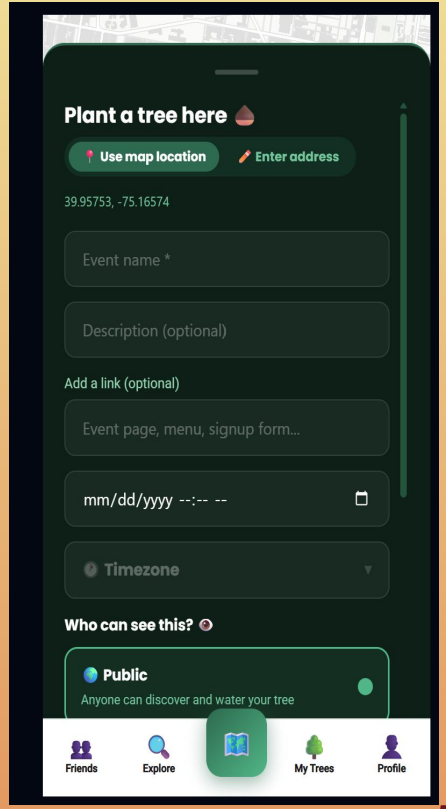
Watering

- Displays the seeds you have watered
- See other who watered and supported

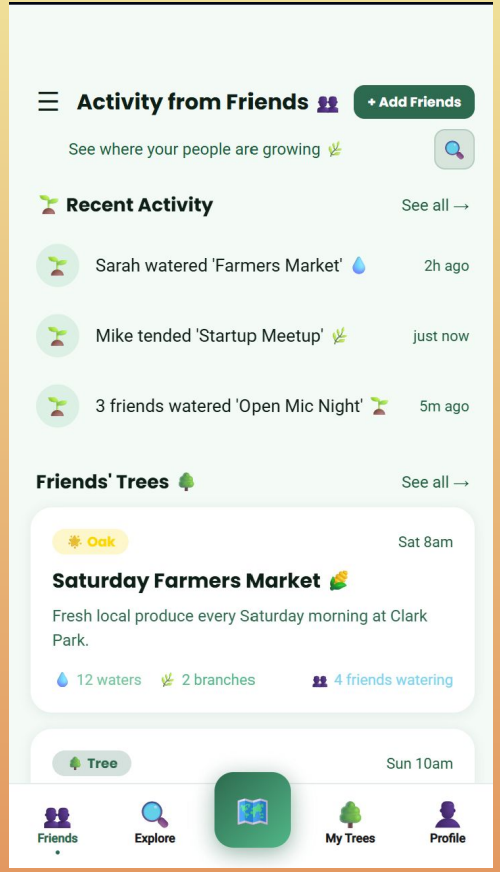
Branches

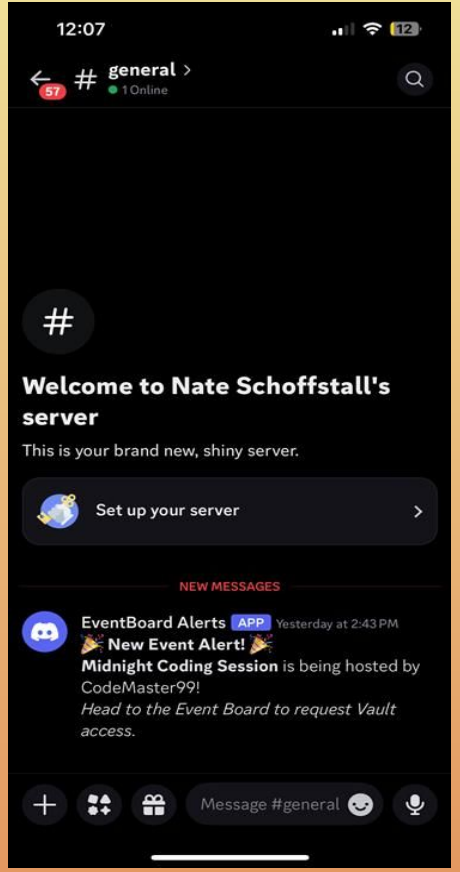
- Sub-events of events.
- E.g. A hot dog eating event at the country fair event.

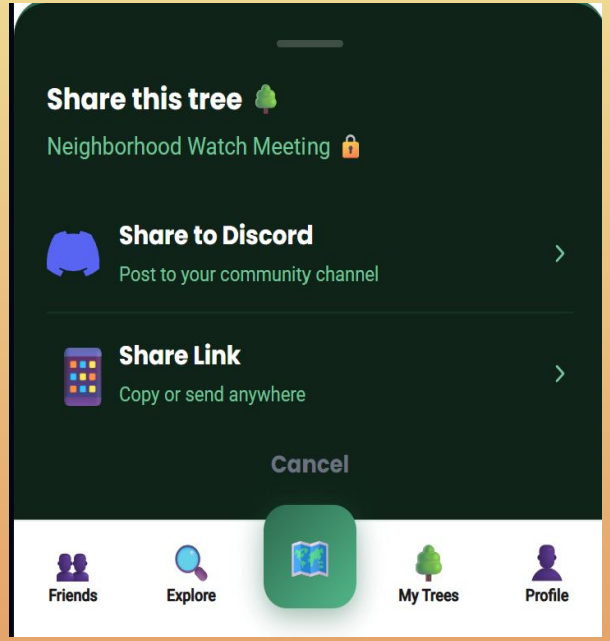




Connect with Friends and Share on Social Media!







Our Backend

+Snowflake Integration

```
spark@docker-1:~$ ls
data  dockerinstall.bash  docker-run.bash  runImage.bash  testsite  updateImage.bash
spark@docker-1:~$ DEBUG=1 bash docker-run.bash
[DEBUG] Step 1: Checking Docker
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
[DEBUG] Step 2: Ensuring .env
[DEBUG] Step 3: Pulling image
Pulling image: ghcr.io/ce-de-ml-nathanschoff/whats:main
[DEBUG] Step 6: Cleaning existing container
Stopping and removing existing container: comunitree
comunitree
comunitree
[DEBUG] Step 7: Running container
Starting Comunitree (frontend 8000, backend 7000)...

Comunitree is running.
  Frontend: http://localhost:8000
  Backend:  http://localhost:7000
  Logs:     docker logs -f comunitree
  Stop:   docker stop comunitree
  Attach: ./docker-run.bash --foreground

If this is the first run and you have not initialized Snowflake yet, run:
  ./docker-run.bash --init-db    # then re-run this script to start the app
  ./docker-run.bash --migrate    # optional: run migrations (after init)
spark@docker-1:~$
```

We connected to our home Research Server in Frostburg, MD to host our site!

Visit now: Sparklabs.frostburg.edu/Communitree

We used Snowflake to Cloud host and connect from our device. Frostburg to the Cloud!

Tech Stack

Frontend

- React + Vite
- Tailwind CSS
- Framer Motion

Maps

- MapLibre GL JS (no API key)

Backend

- Spark Labs Server

Data & APIs

- Supabase
- Snowflake API

What we Learned

Technical

- Scalability and real-time updates are challenging
- Incremental Development
- Integrating SnowFlake API

Project

- Clear architecture saves time
- Simplicity drives better design
- Team communication is key

Future Work

Features

- Expand social features
- Add personalization
- Advanced gamification

Technical

- Improve scalability
- Enhance security
- Optimize database

UX

- Improve onboarding
- Better search & filters
- More growth stages

Thank you for your time!